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Code	<pre> R - R 4.6.1 - ~/Documents/R Programming/NPTEL/ > # --- Lecture 36: Print and Format with Concatenate --- > > # print() handles one object at a time. > # The following gives an error, so it is kept commented: > # print("The zero occurs at", 2*pi, "radians.") > print("The zero occurs at") [1] "The zero occurs at" > print(2*pi) [1] 6.283185 > print("radians") [1] "radians" > # cat() combines multiple items into continuous output > cat("The zero occurs at", 2*pi, "radians.", "\n") The zero occurs at 6.283185 radians. > > d = date() > cat("Today's date is:", d, "\n") Today's date is: Thu Oct 1 14:08:05 2026 > > x = 1:10 > x [1] 1 2 3 4 5 6 7 8 9 10 > cat(x, sep = " ++ ") 1 ++ 2 ++ 3 ++ 4 ++ 5 ++ 6 ++ 7 ++ 8 ++ 9 ++ 10> cat("\n") > > cat(x, sep = " / ") 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10> cat("\n") > > x = 7 > cat("The square of", x, "is", x^2, "\n") The square of 7 is 49 > cat("The square root of", x, "is approximately", + format(sqrt(x), digits=3), "\n") The square root of 7 is approximately 2.65 > > evenno = c(2,4,6,8,10) > evenno [1] 2 4 6 8 10 > cat("The first few even numbers are:", evenno, "\n") The first few even numbers are: 2 4 6 8 10 ... R - R 4.6.1 - ~/Documents/R Programming/NPTEL/ > x = 1:10 > cat(x, fill = 2, labels = paste("C", letters[1:10], "):")) (a): 1 (b): 2 (c): 3 (d): 4 (e): 5 (f): 6 (g): 7 (h): 8 (i): 9 (j): 10 > # --- Lecture 37: paste Function --- > > paste(1:12) [1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10" "11" "12" > as.character(1:12) [1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10" "11" "12" > > paste("Everybody", "loves", "R Programming.") [1] "Everybody loves R Programming." > paste("Everybody", "loves", "R Programming.", sep="*") [1] "Everybody*loves*R Programming." > paste("Everybody", "loves", "R Programming.", sep="====") [1] "Everybody====loves====R Programming." > > names = c("Prof. Singh", "Mr. Venkat", "Dr. Jha") > names [1] "Prof. Singh" "Mr. Venkat" "Dr. Jha" > paste(names, "is", "a good", "person.") [1] "Prof. Singh is a good person." "Mr. Venkat is a good person." [3] "Dr. Jha is a good person." > paste(names, "is", "a good", "person.", collapse=", and ") [1] "Prof. Singh is a good person., and Mr. Venkat is a good person., and Dr. Jha is a good perso n." > > paste(1, " is first", 2, " is second", 3, " is third", sep="#") [1] "1# is first#2# is second#3# is third" > paste(1, " is first", " ", 2, " is second", " ", + 3, " is third", " ", sep="#") [1] "1# is first #2# is second #3# is third " > > x = paste("Ex", 1:5, sep="_") > x [1] "Ex_1" "Ex_2" "Ex_3" "Ex_4" "Ex_5" </pre>
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Console | Terminal | Background Jobs
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> x[1]
[1] "Ex_1"
> x[2]
[1] "Ex_2"
> x[3]
[1] "Ex_3"
> x[5]
[1] "Ex_5"
> x = paste("Ex", 1:5, sep="_", collapse="")
> x
[1] "Ex_1Ex_2Ex_3Ex_4Ex_5"
> x[1]
[1] "Ex_1Ex_2Ex_3Ex_4Ex_5"
>
> paste0(1:10)
[1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10"
> paste(1:10)
[1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10"
> paste0(1:10, c("st", "nd", "rd", rep("th", 7)))
[1] "1st" "2nd" "3rd" "4th" "5th" "6th" "7th" "8th" "9th" "10th"
> paste(1:10, c("st", "nd", "rd", rep("th", 7)))
[1] "1 st" "2 nd" "3 rd" "4 th" "5 th" "6 th" "7 th" "8 th" "9 th" "10 th"
> # --- Lecture 38: strsplit Function ---
>
> x = "The&lsyntax&lof&lpaste&lis&lavailable!&inthe online-help"
> x
[1] "The&lsyntax&lof&lpaste&lis&lavailable!&inthe online-help"
> strsplit(x, split="!")
[[1]]
[1] "The&"          "syntax&"          "of&"          "paste&"
[5] "is"            "&avavailable"      "&inthe online-help"
> strsplit(x, split="&")
[[1]]
[1] "The"          "syntax"
[3] "of"           "paste"
[5] "is!&avavailable!&inthe online-help"
>
> y = strsplit(x, split="!&")
> y
[[1]]
[1] "The&lsyntax&lof&lpaste&lis" "available"          "inthe online-help"
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Console Terminal Background Jobs
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> y[[1]][1]
[1] "The&isyntax&lof&lpaste&lis"
> y[[1]][2]
[1] "available"
> y[[1]][3]
[1] "inthe online-help"
>
> dates = c("2020-07-24", "2021-08-25", "2022-09-26", "2023-10-27")
> datesplt = strsplit(dates, "-")
> datesplt
[[1]]
[1] "2020" "07" "24"

[[2]]
[1] "2021" "08" "25"

[[3]]
[1] "2022" "09" "26"

[[4]]
[1] "2023" "10" "27"

>
> datemat = matrix(unlist(datesplt), nrow=4, ncol=3, byrow=TRUE)
> datemat
      [,1] [,2] [,3]
[1,] "2020" "07" "24"
[2,] "2021" "08" "25"
[3,] "2022" "09" "26"
[4,] "2023" "10" "27"
>
> datematrix = matrix(as.numeric(unlist(datesplt)),
+                      nrow=4, ncol=3, byrow=TRUE)
> datematrix
      [,1] [,2] [,3]
[1,] 2020  7 24
[2,] 2021  8 25
[3,] 2022  9 26
[4,] 2023 10 27
>
> strsplit("Shalabh", split="")
[[1]]
[1] "S" "h" "a" "l" "a" "b" "h"
>
>

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> # --- Lecture 39: Manipulations with Strings and Alphabets ---
>
> x = "R course 24.07.2022"
> y = "Number of participants: 25"
> nchar(x)
[1] 19
> nchar(y)
[1] 26
>
> nzchar(x)
[1] TRUE
> nzchar(y)
[1] TRUE
> x = c("Apple", "Banana", "Cake")
> x
[1] "Apple" "Banana" "Cake"
> nzchar(x)
[1] TRUE TRUE TRUE
>
> y = c("Apple", "", "Cake")
> y
[1] "Apple" "" "Cake"
> nzchar(y)
[1] TRUE FALSE TRUE
>
> x = c("Apple", "Banana", "Cake")
> nchar(x)
[1] 5 6 4
>
> y = c(2, 4, 6)
> nchar(y)
[1] 1 1 1
>
> z = c(11, 222, 3333)
> nchar(z)
[1] 2 3 4
>
> z1 = c(1.1, 2.22, 3.333)
> nchar(z1)
[1] 3 4 5
>
> x = "R course will start from 24.07.2022"
> toupper(x)
[1] "R COURSE WILL START FROM 24.07.2022"
>

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>
> z = "INDIAN INSTITUTE OF TECHNOLOGY"
> tolower(z)
[1] "indian institute of technology"
>
```

Environment			History	Connections	Git	Tutorial
R - Global Environment			Import Dataset	59 MIB		
Data						
ab	List of 6					
datemat	chr [1:4, 1:3] "2020" "2021" "2022" "2023" "07" "08" "09" "10" "2...					
datematrix	num [1:4, 1:3] 2020 2021 2022 2023 7 ...					
datesplt	List of 4					
list1	List of 6					
list12	List of 6					
list2	List of 6					
Values						
brands	chr [1:7] "A" "A" "B" "B" "B" "B" "C"					
brands_fac	Factor w/ 3 levels "A","B","C": 1 1 2 2 2 2 3					
colours	chr [1:3] "blue" "green" "red"					
d	"Thu Oct 1 14:08:05 2026"					
dates	chr [1:4] "2020-07-24" "2021-08-25" "2022-09-26" "2023-10-27"					
evenno	num [1:5] 2 4 6 8 10					
facy	Factor w/ 6 levels "one","two","three",...: 1 4 3 5 4 2 4					
income	Ord.factor w/ 3 levels "low"<"medium"<...: 3 3 1 2 2					
labels.dieface	chr [1:6] "one" "two" "three" "four" "five" "six"					
names	chr [1:3] "Prof. Singh" "Mr. Venkat" "Dr. Jha"					
possible.dieface	num [1:6] 1 2 3 4 5 6					
x	"R course will start from 24.07.2022"					
y	num [1:3] 2 4 6					
z	"INDIAN INSTITUTE OF TECHNOLOGY"					
z1	num [1:3] 1.1 2.22 3.33					